Performance paradoxes in matching systems are not that rare

A. Busic[1], J.M. Fourneau[1,2], A. Lunven[1], S. Li[1]

1, INRIA ARGO, Paris, France
ana.busic@inria.fr, antoine.lunven@ens-paris-saclay.fr, shu.li@inria.fr
2 DAVID, Univ. Paris-Saclay, UVSQ, Versailles France
Jean-Michel.Fourneau@uvsq.fr

Abstract. A Matching model describes the waiting times suffered by items before they match with other items and disappear immediately without service. The matching relation is described by a compatibility graph. It is an easy representation of multiple types synchronizations between items. When we add an edge in the compatibility graph, one expect that the expected total number of items decreases. Unexpectedly this is not always true and there is a performance paradox. Extending previous results we show new matching models with performance paradoxes which seems to be more frequent than one can expect.

1 Introduction

Following [1] a Matching model is a triple (G, Φ, α) defined by

1. a matching graph $G = (\mathcal{V}, \mathcal{E})$ which is an undirected graph whose nodes in $\mathcal{V}$ are classes of items and whose edges in $\mathcal{E}$ models the allowed matching of items. G is called the compatibility graph or the Matching graph. Upon arrival, an item is queued if there are not compatible items present in the system.
2. Φ is a matching policy. It states the couple of items which is chosen upon arrival when an arriving item matches one or more items already waiting. Typical matching disciplines are First Come First Match (FCFM, an analog of First Come First Served in this approach) or Match the Longest Queue. Once they are matched, both items leave the system immediately (no need for service).
3. a distribution of probability α to model the arrivals of items. Alternatively one can consider a collection of Poisson processes for a continuous-time model.

A matching occurs when two (or more) compatible items are present and it is performed according to the matching discipline.

The general matching model proposed in [2] and [1] was considering a general undirected matching graph G and it is assumed that the arrivals of items occur one at a time. It is important to avoid the confusion with Bipartite Matching

Model (see for instance [3] and references therein) where the matching graph is bipartite and two items of distinct classes arrive at the same time.

General matching models are easy to describe but they not so simple to analyze. Assuming independent Poisson arrivals of items, and FCFM discipline the model is associated with an infinite Markov chain. A necessary condition of stability and a product form solution were proved in [2] and [1] for these assumptions.

Some of us have recently established that there exists some performance paradox for FCFM matching models [4]. When one add new edges in the compatibility graph, one may expect that the expectation of the total number of customers decreases. In [4], some examples were presented to show that it is not always the case and a sufficient condition for such a performance paradox to exist is proved. Thus, adding flexibility on the matching does not always result in a performance improvement. The example was based on an almost complete graph. In [5], the results have extended in many directions: all greedy disciplines instead of FCFM, and an infinite family of compatibility graphs. However some questions remain about the compatibility graphs which lead to performance paradoxes.

An application of Matching models studied in the literature was the kidney exchanges [6,7,8]. The kidney exchange arises when a healthy person who wishes to donate a kidney is not compatible (blood types or tissue types) with the receiver. Two incompatible pairs (or maybe more) can form a cyclic exchange, so that each patient can receive a kidney from a compatible donor.

The technical part of the paper is as follows. We begin in the next section with a short presentation of performance paradoxes for Matching systems. Section 3 is devoted to a simple example built for a compatibility graph with 5 nodes which was not considered previously. In Section 4, we also find some performance paradoxes for multi-graphs.The objective of the paper is to show arrival rates and compatibility graphs which exhibit performance paradoxes. The vector of arrival rates have been obtained through brute force inspection. Two numerical software based on the closed form solution for expectation of total number of items based on the independent sets have been developed for cross-validation and a simulator was also used during the validation process.

2 Performance Paradoxes in Matching Systems

Let $G = (\mathcal{V}, \mathcal{F})$ be the matching graph. Usually, nodes in $\mathcal{V}$ are also denoted as letters and an ordered list of letters is called a word. Assume that x is a letter from $\mathcal{V}$, $\Gamma(x)$ is the set of neighbors of x in G. In this paper, we only consider FCFM matching disciplines even if some results on paradoxes have been established for the larger set of greedy disciplines [5]. In this paper, $\mathbb{E}[G, \alpha]$ will denote the expected number of items for compatibility graph G and vector of arrival rates α. Let G be a graph and a, b two nodes of V such that (a, b) is not an edge of G, we will denote as $G + e$, the graph G where we have added edge $e = (a, b)$. Similarly, $G - (u, v)$ where (u, v) is an edge of G, is the graph

obtained after removing edge (u, v). Finally, $\mathcal{I}$ will denote the set of Independent Sets of the compatibility graph.

Theorem 1 (Product form for the steady-state distribution). *[2] Assuming ergodicity, the steady state distribution for word w is:*

$$\pi(w) = C \frac{\prod_{i=1}^{l(w)} \alpha_{w_i}}{\prod_{k=1}^{l(w)} \alpha(\Gamma(\{w_1, ..w_k\}))},$$

where $l(w)$ is the length of word w, w_i is the i-th letter of w, $\alpha(S)$ is the sum of the arrival rates for letters in S.

But it remains to compute the normalizing constant (i.e. C) and the expectation of the total number of items in the system: a difficult task as the state space of the Markov chain is infinite. Hopefully, close-form expressions have been obtained for these quantities.

Corollary 1 (Expectation) *[4] For all independent sets $\mathcal{I} \in \mathbb{I}$, consider an ordered version of $\mathcal{I}$ noted $\mathcal{I}^o = \left\{i_1, \cdots, i_{|\mathcal{I}|}\right\}$, we note σ a permutation of its elements, i.e $\mathcal{I}^{\sigma(o)} = \left\{i_{\sigma(1)}, \cdots, i_{\sigma(|\mathcal{I}|)}\right\}$ and $\mathfrak{S}_{|\mathcal{I}|}$ the set of all permutations of $[\![1, |\mathcal{I}|]\!]$. We define*

$$T_{\mathcal{I}^o} = \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha_{i_k}}{|\alpha(\Gamma(\{i_1, \cdots, i_k\}))| - |\alpha(\{i_1, \cdots, i_k\})|}$$

and $T_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} T_{\mathcal{I}^{\sigma(o)}}$. We also define:

$$E_{\mathcal{I}^o} = \sum_{l=1}^{|\mathcal{I}|} \frac{|\alpha(\Gamma(\{i_1, \cdots, i_l\})|}{|\alpha(\Gamma(\{i_1, \cdots, i_l\}))| - |\alpha(\{i_1, \cdots, i_l\})|} \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha_{i_k}}{|\alpha(\Gamma(\{i_1, \cdots, i_k\}))| - |\alpha(\{i_1, \cdots, i_k\})|}$$

and $E_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} E_{\mathcal{I}^{\sigma(o)}}$. $\mathbb{E}[G, \alpha]$ can be rewritten as a finite sum over all independent sets, i.e

$$\mathbb{E}[G, \alpha] = \left(1 + \sum_{\mathcal{I} \in \mathbb{I}} T_{\mathcal{I}}\right)^{-1} \left(\sum_{\mathcal{I} \in \mathbb{I}} E_{\mathcal{I}}\right) \tag{1}$$

Computing the expectation is based on two summations: the inner one on the independent sets and the outer one on the permutation of the letters of the independent sets. Remark that as the number of independent sets can be exponential on $|\mathcal{V}|$, such an algorithm is in practice limited to compatibility graph with a small number of nodes.

Definition 1 (Performance Paradox) *For some arrival rates α, we have a performance paradox, if $\mathbb{E}[G + e, \alpha] > \mathbb{E}[G, \alpha]$.*

Property 1 (First Performance Paradox [4]) *Let K_4 be the complete graph with 4 nodes. For some arrival rates α, we have stated that*

$$\mathbb{E}[K_4 - (1,2), \alpha] > \mathbb{E}[K_4, \alpha].$$

The analysis in [4] requires one combinatorial notion. The bottleneck set, denoted $BS(G, \alpha)$ throughout the paper, is defined as

$$BS(G, \alpha) = \{I \in \mathcal{I} \mid argmin(\alpha(\Gamma(I)) - \alpha(I))\}.$$

Remember that a necessary and sufficient condition for stability under the FCFM policy is that $\alpha(\mathcal{I}) < \alpha(\Gamma(\mathcal{I}))$ for every independent set $\mathcal{I}$.

As it is difficult to use the formula on the expectation, we have developed in [5] some "lifting theorems" to explain how one can build large compatibility graphs and arrival rates based on small compatibility graphs. We have proved that the existence of performance paradoxes on small graphs implies the existence of such paradoxes on large graphs. Then from the initial seed graph (i.e. $K_4 - (1,2)$ in [4]) we have obtained an infinite set of graphs with paradoxes with the help of the JOIN operation, the definition of which follows.

Definition 2 (JOIN $\bowtie$) *. Let $G1 = (V1, E1)$ and $G2 = (V2, E2)$ two arbitrary graphs, the join of $G1$ and $G2$ (denoted as $G1 \bowtie G2$) is defined by the set of nodes $V1 \cup V2$ and the set of edges $E1 \cup E2 \cup V1 \times V2$, where $\times$ is the Cartesian product.*

Property 2 (Construction of Performance Paradox [5]) *Performance paradoxes exist for graphs $IN_n \bowtie (K_4 - (1,2))$ where IN_n is the graph with n isolated nodes.*

From these results, it is clear that the graphs exhibiting paradoxes are not necessarily dense because the average degree may be as small as 4. And due to the constructions, all the graphs with paradoxes we have found contain a star. Thus, in the next section, we work on a small exemples of graph which does not contain a star and the average degree of which is 1.

3 Star-free Graphs

We say that a graph G is *star-free* if G contains no induced subgraph isomorphic to a star. A routine extremal argument shows that the smallest star-free, non-bipartite graphs are necessarily odd unicyclic graphs on five nodes. A unicyclic graph is a connected graph, where the number of nodes equals to the number of edges; it is odd unicyclic when that unique cycle has odd length. Guided by these facts, we begin with the two odd-unicyclic graphs on 5 nodes: cycle C_5 and F_1 as depicted in Fig. 1. Next, we enlarge the vertex set by one and study F_3, obtained from C_6 by inserting an edge. It is the smallest example in which two distinct odd cycles coexist. Finally, we study the 7-cycle C_7 with 28 independent sets. The arrival rates leading to paradoxes are found by exhaustive search on the sets of parameters which insure the stability.

3.1 Graphs C_5, F_1 and F_2

Clearly, graph C_5 has 5 nodes and 5 edges, it is connected, and the number of independent sets is 10. Thus, it is possible to use the closed form solution to compute $\mathbb{E}[C_5, \alpha]$ for any vector of arrival rates.

The unicyclic graph F_1 shown in Fig. 1 is defined in [9]. By inserting the edges $\{x_2, x_5\}$ in C_5 and $\{z_4, z_5\}$ in F_1, respectively, we obtain a graph isomorphic to the house graph, denoted F_2 ([9, p. 12]). In what follows we analyse cycle graph C_5, unicyclic graph F_1, and house graph F_2.

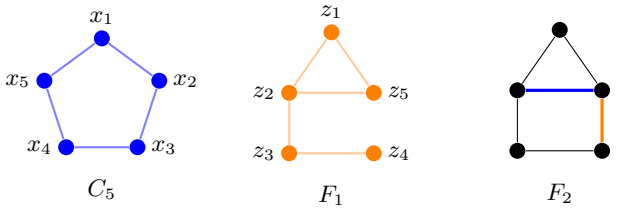


Fig. 1. Graphs: C_5, F_1 and F_2 .

Table 1 summarises the computed values of $\mathbb{E}[G, \alpha]$ for graphs C_5, F_1 and F_2 under distinct arrival-rate vectors α using the Python code in [10] , listed in the first column. For cycle C_5, replacing it by F_2 raises the expectation from 7.86 to 8.17. A similar upward shift is observed when the unicyclic graph F_1 are augmented to F_2 (see Row 3).

Table 1. $\mathbb{E}[G, \alpha]$ for C_5, F_1 and F_2 with arrival rates α in column 1.

α	$\mathbb{E}[G]$	$BS(G, \alpha)$	$\mathbb{E}[G']$	$BS[G', \alpha]$
1: 0.111, 2: 0.10575, 3: 0.179426, 4: 0.35046, 5: 0.253364	7.86	$\{v_1, v_4\}$	8.17	$\{v_1, v_4\}$
1: 0.266025, 2: 0.11005, 3: 0.210855, 4: 0.14145, 5: 0.27162	16.069	$\{v_3, v_5\}$	16.89	$\{v_3, v_5\}$

3.2 F_3: A Star-Free Graph with Two Odd Cycles

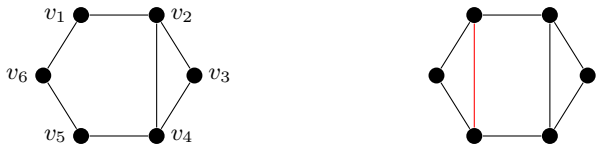


Fig. 2. F_3 and $F_3 + \{v_1, v_5\}$.

As noted previously, graph F_3 is obtained by inserting edge $\{v_2, v_4\}$ into graph C_6. The resulting star-free graph contains two distinct odd cycles with lengths 3 and 5, with 15 independent sets. With rates $\alpha_1 = 0.231787$, $\alpha_2 = 0.462366$, $\alpha_3 = 0.14472$, $\alpha_4 = 0.127484$, $\alpha_5 = 0.00342$ and $\alpha_6 = 0.030223$, we get with the Python code described in [10] an expectation of total number of items equal to 39.684 for graph F_3. This expectation increases to 39.697 when we add edge $\{v_1, v_5\}$, again a performance paradox. Both graphs have bottleneck independent set equal to $\{v_2, v_6\}$.

3.3 C_7: Performance Paradox on a Simple Cycle

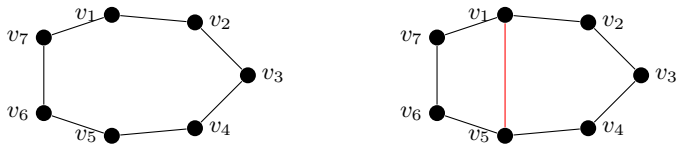


Fig. 3. C_7 and $C_7 + \{v_1, v_5\}$.

Graph C_7 is a simple odd cycle, with 28 independent sets. We study cycle graph C_7 together with graph $C_7 + \{v_1, v_5\}$, as illustrated in Fig. 3. Using the Python code in [10], we select the arrival rate vector

$$\alpha = [0.045206, 0.240384, 0.303789, 0.146114, 0.132526, 0.111772, 0.020209].$$

For this choice of parameters we obtain $\mathbb{E}[C_7, \alpha] = 268.29593$ and $\mathbb{E}[C_7 + \{v_1, v_5\}, \alpha] = 285.302751$. Both graphs share the same bottleneck set $\{v_2, v_4.v_6\}$.

4 (Multi)-Graph with self loops

Here, we assume that some nodes in the matching graph have a self-loop and this was clearly forbidden by the previous assumptions in [1]. Thus the compatibility graph becomes a multigraph. Note that the results obtained in [2] and [1] are not valid anymore because of some technical details in the proofs of product form. Such a model may represent how we can organize fair competition between players with roughly the same ranking (for instance ELO points for chess). The theory for matching models with multigraph has recently been presented in [11] where the authors present a closed form solution for the steady-state distribution when it exists.

4.1 Product form on multigraphs

Let $G = (\mathcal{V}, \mathcal{E})$ be a multigraph with the nodes partition $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ where $\mathcal{V}_1 = \{i \in \mathcal{V} \mid \{i, i\} \in \mathcal{E}\}$ and $\mathcal{V}_2 = \mathcal{V} \setminus \mathcal{V}_1$. Denote by $\mathbb{I}(\check{G})$ the family of independent sets of $\check{G} = (\mathcal{V}, \mathcal{E} \setminus \{(i, i) \mid i \in \mathcal{V}_1\})$.

Theorem 2 (Product form [11, Theorem 4.5]) *Assume the FCFM policy and ergodicity. For any word $w = (w_1, \ldots, w_q) \in W$,*

$$\pi(w) = \pi(\emptyset)\bar{\pi}(w) = \pi(\emptyset) \prod_{l=1}^{q} \frac{\alpha(w_l)}{\alpha\big(\Gamma(\{w_1, \ldots, w_l\})\big)}, \tag{2}$$

where

$$\pi(\emptyset)^{-1} = 1 + \sum_{\mathcal{I} \in \mathbb{I}(\check{G})} \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(e_{\sigma(k)})}{\alpha\big(\Gamma(\{e_{\sigma(1)}, \ldots, e_{\sigma(k)}\})\big) - \alpha\big(\{e_{\sigma(1)}, \ldots, e_{\sigma(k)}\} \cap \mathcal{V}_2\big)}.$$

Ref [11] does not provide a closed form solution for the expectation. Thus we develop such a formula in the following. For $\mathcal{I} \in \mathbb{I}(\check{G})$, fix an arbitrary ordering $\mathcal{I}^o = \{e_1, \ldots, e_{|\mathcal{I}|}\}$. Given a permutation σ of the index set $\{1, \ldots, |\mathcal{I}|\}$, define the reordered list $\mathcal{I}^{\sigma(o)} = \{e_{\sigma(1)}, \ldots, e_{\sigma(|\mathcal{I}|)}\}$, that is, the sequence obtained by applying σ to the ordered version $\mathcal{I}^o$. We denote $\mathcal{I}^o_k$ as the first k elements of $\mathcal{I}^o$. Let $\mathcal{J}^o_k = \mathcal{I}^o_k \cap \mathcal{V}_2$. We define

$$T_{\mathcal{I}^o} = \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(e_k)}{\alpha\Big(\Gamma\big(\mathcal{I}^o_k\big)\Big) - \alpha\Big(\mathcal{J}^o_k\Big)},$$

and $T_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} T_{\mathcal{I}^{\sigma(o)}}$. Thus $\pi(\emptyset) = (1 + T_{\mathcal{I}})^{-1}$. We also define

$$E_{\mathcal{I}^o} = \sum_{l=1}^{|\mathcal{I}|} \frac{\alpha\Big(\Gamma\big(\mathcal{I}^o_l\big)\Big)}{\alpha\Big(\Gamma\big(\mathcal{I}^o_l\big)\Big) - \alpha\Big(\mathcal{J}^o_l\Big)} \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(e_k)}{\alpha\Big(\Gamma\big(\mathcal{I}^o_k\big)\Big) - \alpha\Big(\mathcal{I}^o_k\Big)},$$

and $E_{\mathcal{I}} = \sum_{\sigma \in \mathfrak{S}_{|\mathcal{I}|}} E_{\mathcal{I}^{\sigma(o)}}$.

Corollary 3 (Expectation) *The expected number of items in steady state is*

$$\mathbb{E}[G, \alpha] = \Big(1 + \sum_{\mathcal{I} \in \mathbb{I}(\check{G})} T_{\mathcal{I}}\Big)^{-1} \sum_{\mathcal{I} \in \mathbb{I}(\check{G})} E_{\mathcal{I}}. \tag{3}$$

Proof. Clearly,

$$\mathbb{E}[G, \alpha] = \sum_{w \in W} |w|\, \pi(w) = \pi_0 \sum_{w \in W} |w| \overline{\pi}(w) = \pi_0 \sum_{w \neq \emptyset} |w| \overline{\pi}(w).$$

We now rewrite the sum over all words w as a finite sum over independent sets of $\check{G}$. For $\mathcal{I} \in \mathbb{I}(\check{G})$, let $\overline{\pi}_{\mathcal{I}} = \sum_{\substack{w \in W: \\ \text{alphabet}(w) = \mathcal{I}}} |w| \overline{\pi}(w)$, where "alphabet$(w) = \mathcal{I}$" means the set of distinct nodes (classes) appearing in w is exactly $\mathcal{I}$. Then $\mathbb{E}[G, \alpha] = \pi_0 \sum_{\mathcal{I} \in \mathbb{I}(\check{G})} \overline{\pi}_{\mathcal{I}}$. We claim that $\overline{\pi}_{\mathcal{I}} = E_{\mathcal{I}}$ for each independent set $\mathcal{I} \in \mathbb{I}(\check{G})$.

Consider words w whose first occurrences of the distinct letters occur in the order $i_1, \ldots, i_m$. Label the lengths between first occurrences as follows: let p_l^k ($l \in \mathcal{J}_k^o$) be the number of times letter i_l appears between the first occurrence of i_k (exclusive) and the first occurrence of i_{k+1} (exclusive), for $l \leq k$. Let $n_k = \sum_{l=1}^{k} p_l^k$ and $n_0 = 0$. Then any such word w has total length $|w| = |\mathcal{I}| + \sum_{k=1}^{|\mathcal{I}|} n_k$. And the multinomial coefficient is

$$\binom{n_k}{\{p_j^k\}_{j \in \mathcal{J}_k^o}}.$$

Then, we get $\overline{\pi}_{\mathcal{I}^o}$

$$= \prod_{k=1}^{|\mathcal{I}|} \Big(\frac{\alpha(i_k)}{\alpha(\Gamma(\mathcal{I}_k^o))} \Big) \sum_{n_k=0}^{\infty} \sum_{\sum p_j^k = n_k} \prod_{m=1}^{k} \Big(\frac{\alpha(i_m)}{\alpha\big(\Gamma(\mathcal{I}_k^o)\big)} \Big)^{p_m^k} \binom{n_k}{\{p_j^k\}_{j \in \mathcal{J}_k^o}} \Big(|\mathcal{I}| + \sum_{k=1}^{|\mathcal{I}|} n_k \Big)$$

$$= \prod_{k=1}^{|\mathcal{I}|} \Big(\frac{\alpha(i_k)}{\alpha(\Gamma(\mathcal{I}_k^o))} \Big) \sum_{n_k=0}^{\infty} \Big(\frac{\alpha(\mathcal{J}_k^o)}{\alpha\big(\Gamma(\mathcal{I}_k^o)\big)} \Big)^{n_k} \Big(|\mathcal{I}| + \sum_{k=1}^{|\mathcal{I}|} n_k \Big)$$

$$= \sum_{l=1}^{|\mathcal{I}|} \prod_{k=1}^{|\mathcal{I}|} \Big(\frac{\alpha(i_k)}{\alpha(\Gamma(\mathcal{I}_k^o))} \Big) \sum_{n_k=0}^{\infty} \Big(\frac{\alpha(\mathcal{J}_k^o)}{\alpha\big(\Gamma(\mathcal{I}_k^o)\big)} \Big)^{n_k} (n_l + 1)$$

$$= \sum_{l=1}^{|\mathcal{I}|} \Big(\frac{\alpha(i_l)}{\alpha(\Gamma(\mathcal{I}_l^o))} \Big) \sum_{n_l=0}^{\infty} \Big(\frac{\alpha(\mathcal{J}_l^o)}{\alpha(\Gamma(\mathcal{I}_l^o))} \Big)^{n_l} (n_l + 1) \prod_{\substack{k \in [\![1, |\mathcal{I}|]\!] \\ k \neq l}} \Big(\frac{\alpha(i_k)}{\alpha(\Gamma(\mathcal{I}_k^o))} \Big) \sum_{n_k=0}^{\infty} \Big(\frac{\alpha(\mathcal{J}_k^o)}{\alpha(\Gamma(\mathcal{I}_k^o))} \Big)^{n_k}$$

$$= \sum_{l=1}^{|\mathcal{I}|} \Big(\frac{\alpha(i_l)}{\alpha(\Gamma(\mathcal{I}_l^o))} \Big) \Big(\frac{\alpha(\Gamma(\mathcal{I}_l^o))}{\alpha(\Gamma(\mathcal{I}_l^o)) - \alpha(\mathcal{J}_l^o)} \Big)^2 \prod_{\substack{k \in [\![1, |\mathcal{I}|]\!] \\ k \neq l}} \Big(\frac{\alpha(i_k)}{\alpha(\Gamma(\mathcal{I}_k^o))} \Big) \Big(\frac{\alpha(\Gamma(\mathcal{I}_k^o))}{\alpha(\Gamma(\mathcal{I}_k^o)) - \alpha(\mathcal{J}_k^o)} \Big)$$

$$= \sum_{l=1}^{|\mathcal{I}|} \frac{\alpha(\Gamma(\mathcal{I}_l^o))}{\alpha(\Gamma(\mathcal{I}_l^o)) - \alpha(\mathcal{J}_l^o)} \prod_{k=1}^{|\mathcal{I}|} \frac{\alpha(i_k)}{\alpha(\Gamma(\mathcal{I}_k^o)) - \alpha(\mathcal{J}_k^o)} = E_{\mathcal{I}^o}$$

Then, we sum over all permutation to get Equation (3).

Remark 1 *Setting $\mathcal{V}_2 = \emptyset$, Theorem 1 together with Corollary 1 are recovered as special instances of Theorem 2 and Corollary 3, respectively.*

4.2 Examples of graphs with 4 or 7 nodes.

We consider $G = (\mathcal{V}, \mathcal{E})$ in Fig. 4, where $\mathcal{V}$ is partitioned into $\mathcal{V}_1 = \{v_2, v_3, v_4\}$ and $\mathcal{V}_2 = \{v_1\}$. The independent sets of $\check{G}$ are $\{\{v_1\}, \{v_2\}, \{v_3\}, \{v_4\}, \{v_2, v_4\}, \{v_2, v_3\}\}$. Graph G_2 is obtained by inserting the edge $\{v_2, v_3\}$ (see Fig. 4). The associated Markov Chains are depicted in Fig. 5. The rates are $\alpha_1 = 0.432685$,$\alpha_2 = 0.005861$, $\alpha_3 = 0.141128$ and $\alpha_4 = 0.420326$. The BS set is a singleton (i.e. $(v1)$. We have a paradox as the expectation slightly increases from 2.944793 to 2.947256 when we add the edge.



Fig. 4. Graphs: G and G_2.

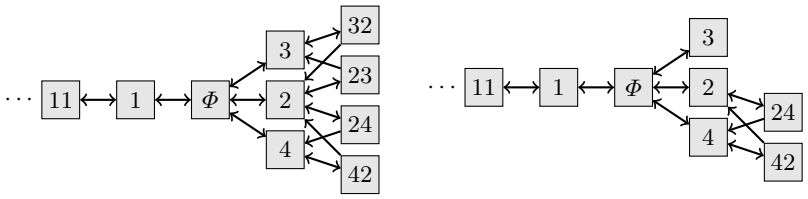


Fig. 5. Truncation of the Markov Chains for G (left) and G_2 (right).

The former example does not address the case of adding a new edge e connecting a node lacking a self-loop to a node with self-loop. To fill this gap, we study graph H_2 (see Fig.6), obtained from H by inserting edge $\{v_1, v_5\}$. Similarly, graph H_1 is formed by connecting v_2 and v_3, while graph H_3 is obtained by adding a self-loop to v_5. We have found performance paradoxes in all cases. For arrival rate vector $(0.070716, 0.180906, 0.238414, 0.204521, 0.164007, 0.07243, 0.069006)$ the bottleneck set is $\{v_7\}$. The expectation is 31.218542 for graph H and it increases to 31.539906 for graph H_1. Similarly, if we start with arrival rate vector $(0.08502, 0.06418, 0.076071, 0.096493, 0.336406, 0.326572, 0.015258)$ for H, we get an expectation for the total number equal to 13.0179 and the expectation is equal to 13.058815 for the same rates and graph H_2. Finally, the comparison between H and H_3 for rate vector $(0.104627, 0.052725, 0.194586, 0.190934, 0.117413, 0.225466, 0.114249)$ also shows a performance paradox: the expectation was 33.838547 for H and it jumps to 35.972599 for H_3. In both cases, the bottleneck independent set is singleton $\{v_6\}$.

5 Conclusions and Remarks

This study extends the catalogue of performance paradoxes in stochastic matching models by exhibiting new star-free, sparsely connected graphs—some with self-loops—where adding a single compatibility edge or loop increases the steady-state number of waiting items. These results demonstrate that the paradox is structurally robust, arising even in small odd-cycle graphs such as F_3 and C_7. Beyond enriching the theoretical landscape, the findings caution practitioners that naive flexibility enhancements may backfire in real-world platforms. Future work should (i) characterise exact arrival-rate conditions for paradox emergence,

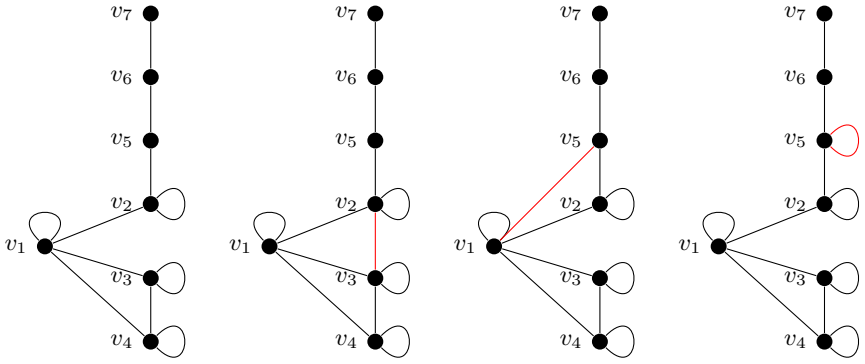


Fig. 6. graphs: H, H_1, H_2 and H_3.

and (ii) explore policy modifications that mitigate such counter-intuitive degradations.

References

1. Jean Mairesse and Pascal Moyal. Stability of the stochastic matching model. *Journal of Applied Probability*, 53(4):1064–1077, December 2016.
2. Pascal Moyal, Ana Bušić, and Jean Mairesse. A product form for the general stochastic matching model. *Journal of Applied Probability*, 58(2):449–468, June 2021.
3. Ana Bušić, Varun Gupta, and Jean Mairesse. Stability of the Bipartite Matching Model. *Advances in Applied Probability*, 45(2):351–378, June 2013.
4. Arnaud Cadas, Josu Doncel, Jean-Michel Fourneau, and Ana Bušić. Flexibility can hurt dynamic matching system performance. *SIGMETRICS Perform. Eval. Rev.*, 49(3):37–42, March 2022.
5. Ana Bušić, Arnaud Cadas, Josu Doncel, and Jean-Michel Fourneau. Performance paradox of dynamic matching models under greedy policies. *Queueing Systems*, 107(3-4):257–293, September 2024.
6. M. Utku Ünver. Dynamic Kidney Exchange. *The Review of Economic Studies*, 77(1):372–414, January 2010.
7. United Network for Organ Sharing. Living donation kidney paired, Online. `https://unos.org/wp-content/uploads/unos/livingdonationkidneypaired.pdf`.
8. Itai Ashlagi, Patrick Jaillet, and Vahideh H. Manshadi. Kidney exchange in dynamic sparse heterogenous pools. In *Proceedings of the Fourteenth ACM Conference on Electronic Commerce*, EC '13, pages 25–26, New York, NY, USA, June 2013. Association for Computing Machinery.
9. Douglas Brent West. *Introduction to Graph Theory*. Prentice Hall, Upper Saddle River, 2001.
10. Shu Li. Matching Paradox Simulator, May 2025. `https://github.com/shuligraph/matching-paradox-sim`.
11. Jocelyn Begeot, Irène Marcovici, Pascal Moyal, and Youssef Rahme. A general stochastic matching model on multigraphs. *Latin American Journal of Probability and Mathematical Statistics*, 18(1):1325, 2021.